

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03/08/2026	Duration:	60 minutes

Code of Conduct

I Y. Ramadevi - 192373065 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **Y. Ramadevi**

Title: Cloud-Based Hospital Management System Using Microservices Architecture

Problem Statement

A Cloud-Based Hospital Management System (HMS) is designed to digitize and automate the daily operations of hospitals using Microservices Architecture deployed on cloud infrastructure. The system enables patients to register online, book appointments, access medical records, pay bills, and receive notifications. Doctors can manage appointments, diagnose patients, prescribe medicines, and access electronic health records (EHR). Hospital administrators manage departments, staff, billing, inventory, reports, and system configurations.

The system consists of independent microservices such as Patient Service, Doctor Service, Appointment Service, Billing Service, Pharmacy Service, Laboratory Service, Notification Service, Authentication Service, and Reporting Service. These services communicate through REST APIs and asynchronous messaging, ensuring high scalability, reliability, and maintainability. Cloud deployment enables automatic scaling, disaster recovery, and high availability, making the system suitable for hospitals with multiple branches and thousands of concurrent users.

1. Analyse System Requirements

Study the given problem statement and identify the following:

System Objectives

- Develop a centralized cloud-based hospital management platform.
- Automate patient registration, appointment scheduling, billing, pharmacy, and laboratory management.
- Improve healthcare service quality through real-time information sharing.
- Ensure secure storage and retrieval of electronic medical records.
- Support multiple hospitals and branches through cloud deployment.
- Enable remote access for doctors, patients, and administrators.

Functional Requirements

- Patient Registration and Login
- Doctor Registration and Login
- Appointment Scheduling
- Patient Medical Records Management
- Electronic Health Record (EHR) Management
- Doctor Consultation Management
- Pharmacy Management
- Laboratory Test Management
- Billing and Payment Management
- Staff Management
- Notification Service

- Report Generation
- User Authentication and Authorization
- Hospital Administration

Non-Functional Requirements

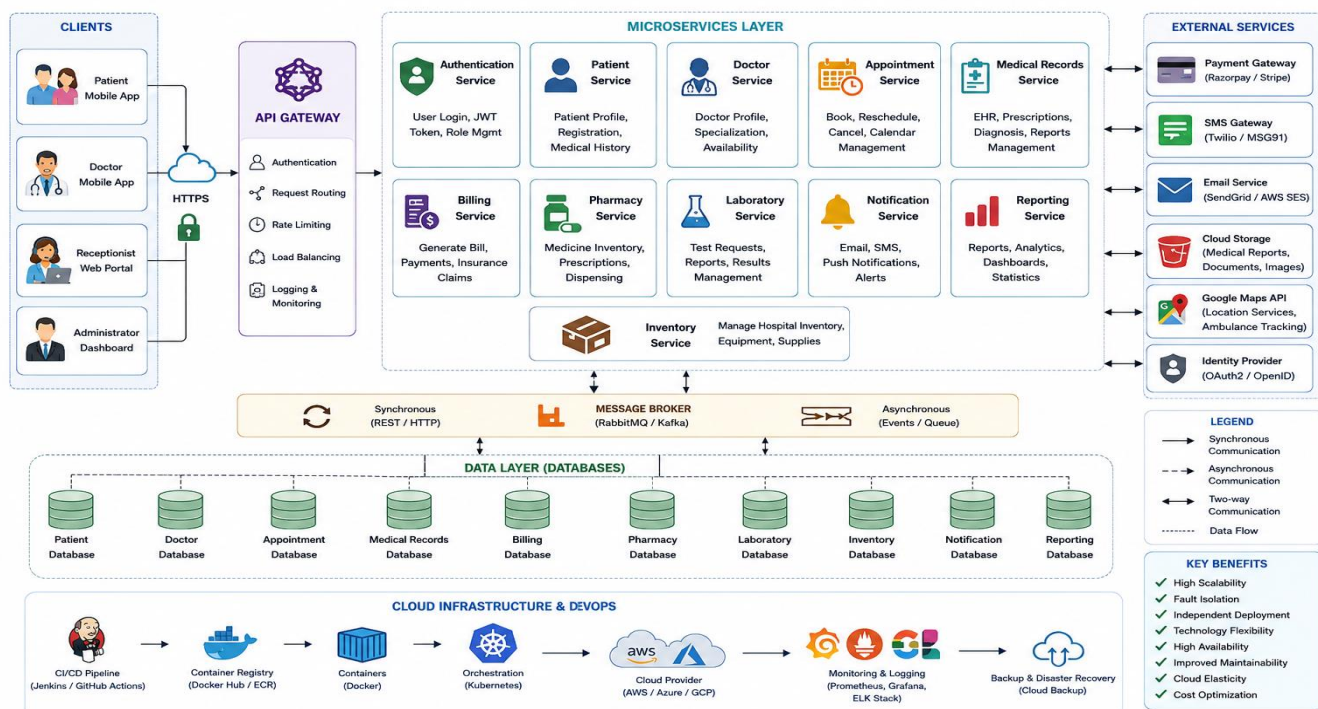
- High Performance
- Scalability
- Security
- Reliability
- Availability
- Maintainability
- Interoperability
- Fault Tolerance
- Data Integrity
- Disaster Recovery

Quality Attributes

Identify and explain the following quality attributes:

- Scalability
- Maintainability
- Reliability
- Availability
- Performance
- Security

Cloud-Based Hospital Management System Using Microservices Architecture



2. Select a Suitable Software Architecture

Selected Architecture

Microservices Architecture

Justification

Microservices Architecture is the most suitable architecture because:

- Each hospital module operates independently.
 - Individual services can be deployed separately.
 - Easy maintenance and debugging.
 - Supports cloud-native deployment.
 - Independent scaling of services.
 - Better fault isolation.
 - Faster software updates.
 - Easier integration with third-party healthcare services.
 - Supports continuous deployment using DevOps.
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3. Draw the Software Architecture Diagram

Design a software architecture diagram showing the following components.

Client Layer

- Patient Mobile Application
- Doctor Mobile Application
- Receptionist Web Portal
- Administrator Dashboard

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API Gateway

- Authentication
- Request Routing
- Rate Limiting
- Load Balancing

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Microservices Layer

- Authentication Service

- Patient Service
- Doctor Service
- Appointment Service
- Medical Records Service
- Billing Service
- Pharmacy Service
- Laboratory Service
- Notification Service
- Reporting Service
- Inventory Service

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Communication Layer

- REST APIs
- RabbitMQ / Kafka Message Broker

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Database Layer

- Patient Database
- Doctor Database
- Billing Database
- Medical Records Database
- Pharmacy Database
- Laboratory Database

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External Services

- Payment Gateway
- SMS Gateway
- Email Service
- Cloud Storage
- Google Maps API (Ambulance Tracking)

4. Explain Components and Their Interactions

Client Layer

Patients, doctors, receptionists, and administrators access the system using web or mobile applications.

API Gateway

The API Gateway authenticates users, validates requests, and routes them to the appropriate microservice.

Authentication Service

Handles user login, registration, JWT token generation, and role-based authorization.

Patient Service

Stores patient details, medical history, insurance information, and appointment records.

Doctor Service

Maintains doctor profiles, schedules, specialties, and consultation details.

Appointment Service

Schedules appointments, checks doctor availability, and manages cancellations and rescheduling.

Medical Records Service

Stores electronic health records, prescriptions, diagnoses, and treatment history securely.

Billing Service

Generates bills, processes payments, manages insurance claims, and maintains financial transactions.

Pharmacy Service

Manages medicine inventory, prescriptions, stock updates, and medicine dispensing.

Laboratory Service

Handles lab test requests, uploads reports, and shares results with doctors and patients.

Notification Service

Sends appointment reminders, prescription alerts, payment confirmations, SMS, and email notifications.

Reporting Service

Generates hospital performance reports, revenue analysis, patient statistics, and doctor workload reports.

Communication Between Components

- Clients communicate with the API Gateway.
 - API Gateway forwards requests to appropriate microservices.
 - Microservices communicate through REST APIs.
 - Events are exchanged asynchronously using RabbitMQ or Kafka.
 - Each service manages its own independent database.
 - External services process payments and notifications.
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5. Discuss Quality Attributes

Scalability

Each microservice scales independently according to demand using cloud auto-scaling.

Security

- OAuth 2.0 Authentication
 - JWT Tokens
 - HTTPS Encryption
 - Role-Based Access Control
 - Encrypted Medical Records
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Performance

- Independent service execution
 - Redis Caching
 - Load Balancing
 - Optimized Database Queries
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Reliability

- Fault isolation
 - Automatic retries
 - Database replication
 - Backup and recovery
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Maintainability

- Independent codebases
 - Modular development
 - Easy testing and deployment
 - Continuous Integration/Continuous Deployment (CI/CD)
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Availability

- Cloud deployment across multiple availability zones
 - Auto failover
 - 24×7 system availability
 - Disaster recovery support
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6. Justify Architectural Decisions

Architectural Rationale

Microservices Architecture was selected because it offers:

- Independent deployment of hospital modules
 - Faster software development
 - Better scalability
 - High availability
 - Easy cloud migration
 - Simplified maintenance
 - Improved fault tolerance
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Trade-Offs

Advantages

- Highly scalable
- Better fault isolation
- Easy maintenance
- Independent deployment
- Technology flexibility
- Continuous deployment support

Disadvantages

- Increased system complexity
 - Higher deployment cost
 - Distributed data management
 - Network communication overhead
 - More monitoring and logging requirements
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Future Enhancements

The proposed architecture supports future enhancements such as:

- AI-based disease prediction
 - Telemedicine and video consultations
 - IoT-enabled patient monitoring
 - Wearable health device integration
 - Blockchain for secure medical records
 - Multi-hospital integration
 - Cloud analytics and business intelligence dashboards
 - Mobile health applications
 - Voice-assisted healthcare services
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Conclusion

The Cloud-Based Hospital Management System Using Microservices Architecture provides a scalable, secure, reliable, and maintainable solution for modern healthcare organizations. By decomposing the system into independent microservices, the architecture enables efficient resource utilization, rapid deployment, fault isolation, and seamless integration with cloud services and external healthcare systems. This architecture supports future technological advancements while ensuring high performance, data security, and continuous availability, making it an ideal solution for large-scale hospital environments.